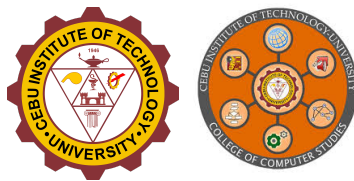


CEBU INSTITUTE OF TECHNOLOGY UNIVERSITY

COLLEGE OF COMPUTER STUDIES



Software Requirements Specification *for* **ELEMENTOPIA**

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2526-sem2-it332-53

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Change History

Date	Version	Description
May 6, 2026	1.0	Initial creation of SRS document.
May 13, 2026	1.1	Creating the introduction and overall description sections.
May 14, 2026	1.2	Creating the specific requirements section and diagrams.
May 21, 2026	1.3	Addition of other diagrams

Table of Contents

Change History	2
Table of Contents	3
1. Introduction	4
1.1. Purpose	4
1.2. Scope	4
1.3. Definitions, Acronyms and Abbreviations	4
1.4. References	4
2. Overall Description	5
2.1. Product perspective	5
2.2. User characteristics	5
2.4. Constraints	5
2.5. Assumptions and dependencies	6
3. Specific Requirements	7
3.1. External interface requirements	7
3.1.1. <i>Hardware interfaces</i>	7
3.1.2. <i>Software interfaces</i>	7
3.1.3. <i>Communications interfaces</i>	7
3.2. Functional requirements	7
<i>Module 1</i>	7
<i>Module 2</i>	8
3.4. Non-functional requirements	8
<i>Performance</i>	8
<i>Security</i>	8
<i>Reliability</i>	8

1. Introduction

1.1. Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a comprehensive and detailed description of the three new modules introduced to the Elementopia gamified chemistry learning platform. This document outlines the functional and non-functional requirements, design constraints, system interfaces, and assumptions that govern the development of the Elemental Resonance Puzzle Module, the Assessment and Progress Module, and the Opponent-Based Challenge Module. It serves as the authoritative reference for developers, testers, project evaluators, and all relevant stakeholders throughout the development lifecycle.

This project extends an existing Elementopia system inherited from a prior development team. The three new modules are built on top of the original platform's features and are designed to transform abstract chemistry concepts into interactive, self-directed learning experiences for high school students. The platform replaces quiz-based assessments and binary right-or-wrong feedback with Elemental Resonance Domains that are environmental puzzle rooms where students synthesize chemical compounds to trigger reactions that clear physical obstacles blocking their progression. Session performance data is persisted to Supabase (PostgreSQL) to support device-agnostic progress tracking and real-time multiplayer synchronization.

1.2. Scope

Elementopia is a web-based gamified chemistry education platform targeting high school students. This project extends the existing platform by introducing three new functional modules that shift the educational focus from rote memorization to algorithmic problem-solving and discovery-based learning.

The three new modules are:

- **Elemental Resonance Puzzle Module** — A text-free environmental puzzle system in which students solve chemical bonding challenges by synthesizing compounds within Domain puzzle rooms to trigger environmental reactions that clear physical obstacles blocking their progression. Domains are structured into foundational Element Rooms and advanced Compound Chambers, unlocked sequentially. A minimum of three correct reactions is required to clear each Domain.
- **Assessment and Progress Module** — An automated session performance tracker that logs compound accuracy and completion speed to Supabase within less than 2 seconds per session. A personalized visual Mastery Dashboard renders the student's performance history, highlighting knowledge gaps per Domain to support self-directed learning without teacher intervention.
- **Opponent-Based Challenge Module** — A real-time competitive matchmaking environment using auto-generated 5-digit access codes and temporary session

nicknames. Two students race to synthesize the correct compound first, with match results determined by Speed-to-Compound evaluation logic via Supabase Realtime. Match results are consolidated into the student's Mastery Dashboard.

The platform is fully mobile-responsive and supports both touch-based and desktop interaction. It does not include teacher administration tools or centralized classroom monitoring. The system does not require user registration; students are identified by a self-assigned temporary nickname, and all session data is stored in Supabase.

1.3. Definitions, Acronyms and Abbreviations

- **SRS** — Software Requirements Specification; a document describing the functional and non-functional requirements of a software system.
- **Elementopia** — The proposed gamified web-based chemistry learning platform for high school students, extending an existing inherited system with three new modules.
- **MVP** — Minimum Viable Product; the core functional version of each module at a minimum of 60% functional completion.
- **UI/UX** — User Interface / User Experience; the visual design and interaction quality of the application.
- **Supabase** — An open-source backend platform providing PostgreSQL database storage and real-time data synchronization, used for all session data persistence and multiplayer communication in this system.
- **Elemental Resonance Domain** — A text-free environmental puzzle room in which a student synthesizes chemical compounds to trigger environmental reactions that clear physical obstacles blocking their progression.
- **Elemental Resonance** — The visual reaction animation triggered when a student successfully synthesizes the correct compound within a Domain, dissolving the obstacle and unlocking the path forward.
- **Meaningful Byproduct** -- A contextual, chemistry-based diagnostic message generated by the system for a predefined set of incorrect compound combination attempts, replacing binary "Wrong" notifications with feedback grounded in actual chemical properties.
- **Hazmat Protocol** — An automated failsafe triggered when the system detects five or more failed combination attempts within a 15-second sliding window, visually disabling irrelevant element tiles to narrow the student's focus without revealing the answer.
- **Element Room** — A foundational Domain puzzle room that must be cleared before a student can access Compound Chambers.
- **Compound Chamber** — An advanced Domain puzzle room that requires multi-step

sequential compound synthesis and is only accessible after foundational Element Rooms are cleared.

- **Mastery Dashboard** — A personalized visual interface that renders a student's performance history from Supabase, displaying compound accuracy and completion speed per Domain to support self-directed knowledge gap identification.
- **Speed-to-Compound** — The competitive evaluation metric used in the Opponent-Based Challenge Module that determines the match winner based on who reaches the correct compound synthesis first using the most efficient reaction path.
- **Access Code** — An auto-generated 5-digit alphanumeric code used to create or join an Opponent-Based Challenge room without manual configuration or user registration.
- **Session Nickname** — A temporary, self-assigned identifier entered by the student at the start of each session, used to associate performance data in Supabase without requiring formal authentication.
- **SRL** — Self-Regulated Learning; an educational framework in which learners independently monitor and control their own learning process.
- **Agile** — An iterative software development methodology supporting incremental delivery and continuous improvement.
- **Sprint** — A time-boxed development iteration in Agile methodology, typically lasting two to four weeks.
- **React** — The JavaScript frontend framework used to build the platform's user interface.
- **Spring Boot** — The Java-based backend framework used for API management and server-side logic.
- **Supabase Realtime** — The real-time data synchronization layer provided by Supabase, used to synchronize competitive match state between two players in the Opponent-Based Challenge Module.

1.4. References

1. Elementopia Software Requirements Specification (v1.7, December 2025) — 2425-sem2-it332-g1/g2/g3/g4/g5/g6-16, Cebu Institute of Technology University
2. Supabase Documentation — Supabase Inc., 2026. Available at: <https://supabase.com/docs>
3. React Documentation — Meta Open Source, 2026. Available at: <https://react.dev>
4. Spring Boot Documentation — VMware Inc., 2026. Available at: <https://spring.io/projects/spring-boot>

2. Overall Description

2.1. Product perspective

Elementopia is a web-based application that extends an existing gamified chemistry platform inherited from a prior development team. The three new modules—the Elemental Resonance Puzzle Module, the Assessment and Progress Module, and the Opponent-Based Challenge Module—are built on top of the original platform's features and are developed using React for the frontend, Spring Boot for backend logic and API management, and Supabase (PostgreSQL) for persistent session data storage and real-time multiplayer synchronization.

The system operates within a client-server architecture. The system processes puzzle logic, compound validation, and Domain progression through a combination of React frontend logic and Spring Boot backend API management. Session performance data, including compound accuracy and completion speed, is written to and retrieved from Supabase, making all progress device-agnostic. The Opponent-Based Challenge Module relies on Supabase Realtime to synchronize match state between two competing players in real time.

The platform does not integrate with any external learning management system, school information system, or third-party authentication provider. It fills a specific gap in the science education technology ecosystem by combining three capabilities typically found in isolation: environmental simulation-based puzzle learning, automated behavioral analytics via Supabase, and peer-to-peer competitive challenge, all within a zero-registration, mobile-first environment.

2.2. User characteristics

General Learners (High School Students)

- Enter a temporary session nickname to access the platform without registration or authentication.
- Interact with Elemental Resonance Domain puzzle rooms by synthesizing chemical compounds to trigger environmental reactions that clear physical obstacles blocking their progression.
- Receive Meaningful Byproduct diagnostic feedback for every incorrect compound combination attempt.
- Track personal performance through the visual Mastery Dashboard, which displays compound accuracy and completion speed per Domain retrieved from Supabase.
- Create or join Opponent-Based Challenge rooms using auto-generated 5-digit access codes to compete against peers in real-time Speed-to-Compound evaluation.

2.4. Constraints

The following constraints limit the design and operational deployment of the Elementopia system:

- **Curriculum Scope:** The chemical compound library is restricted to essential high school chemistry interactions, primarily ionic and covalent bonding. The system does not simulate all periodic table reactions and is not designed as an exhaustive chemistry reference.
- **Development Timeline:** The three new modules must be delivered across four Agile sprints. Feature complexity is scoped accordingly, with a minimum viable product threshold of 60% functional completion per module.
- **No Teacher Administration:** The platform intentionally omits centralized classroom monitoring, grade reporting, and instructor configuration tools. Evaluation is limited to session-based behavioral performance metrics logged to Supabase.
- **No User Registration:** The system does not implement any identity verification, OAuth, or third-party authentication. All user sessions are temporary and identified only by a self-assigned nickname. If a student changes their nickname, prior session data will not be associated with their new identity.
- **Network Dependency:** The Opponent-Based Challenge Module and the Assessment and Progress Module require a stable internet connection for Supabase data synchronization and Supabase Realtime communication. Network instability may disrupt match flow, scoring accuracy, and session logging.
- **Touch Device Optimization:** While the platform is designed to be fully mobile-responsive, rendering consistency may vary across different browser engines and operating system versions.
- **Inherited System Dependency:** The three new modules are built on top of the existing Elementopia platform. Any breaking changes to the original system's architecture may require corresponding adjustments to the new modules.

2.5. Assumptions and dependencies

The following assumptions underpin the requirements stated in this document. Changes to any of these factors may require corresponding revisions to the SRS:

- It is assumed that all target users have access to a modern web browser (Google Chrome v110+, Firefox v100+, Safari v15+) that supports standard web application rendering and WebSocket protocol for Supabase Realtime communication.
- It is assumed that devices used by students support touch-based interaction or standard mouse input, and have a minimum screen resolution of 360px width for proper rendering of the puzzle interface.
- It is assumed that a stable internet connection is available during all active sessions, with a recommended minimum bandwidth of 1 Mbps to support Supabase Realtime synchronization during competitive matches.

- It is assumed that the Supabase project instance is properly configured with the required database schema and real-time channels prior to deployment.
- It is assumed that all chemical compound content — element combinations, valid reaction outcomes, and Meaningful Byproduct diagnostic messages — will be curated and encoded into the predefined compound pool by the development team prior to deployment, based on the official high school chemistry curriculum.
- It is assumed that students will use the platform on their own personal devices for the duration of a session. Session data is stored in Supabase under the student's temporary nickname.
- The development of the three new modules depends on the stability of the inherited Elementopia codebase. The frontend depends on React, the backend depends on Spring Boot, and all session data persistence and real-time functionality depends on Supabase.

3. Specific Requirements

3.1. External interface requirements

3.1.1. Hardware interfaces

Elementopia is designed to run on any device equipped with a modern web browser. The system supports both touch-based mobile devices (smartphones and tablets) and desktop or laptop computers with standard mouse and keyboard input. No specialized hardware peripherals are required. The minimum recommended screen resolution is 360px width to ensure proper rendering of the puzzle interface. For the Opponent-Based Challenge Module, a stable network connection (Wi-Fi or mobile data) is required to maintain Supabase Realtime communication between two players.

3.1.2. Software interfaces

Elementopia operates entirely within a standard web browser and does not require installation of any additional software on the user's device. The system interfaces with the following:

- **Web Browser (Chrome v110+, Firefox v100+, Safari v15+)** - The primary runtime environment for all frontend rendering, puzzle interaction, and UI logic built with React.
- **Spring Boot (Java)** - The backend framework responsible for API management, compound validation logic, and server-side processing.
- **Supabase (PostgreSQL)** - Used to store and retrieve all session performance data, including compound accuracy and completion speed per Domain.
- **Supabase Realtime** - Used exclusively by the Opponent-Based Challenge Module to synchronize match state between two competing players in real time.

3.1.3. Communications interfaces

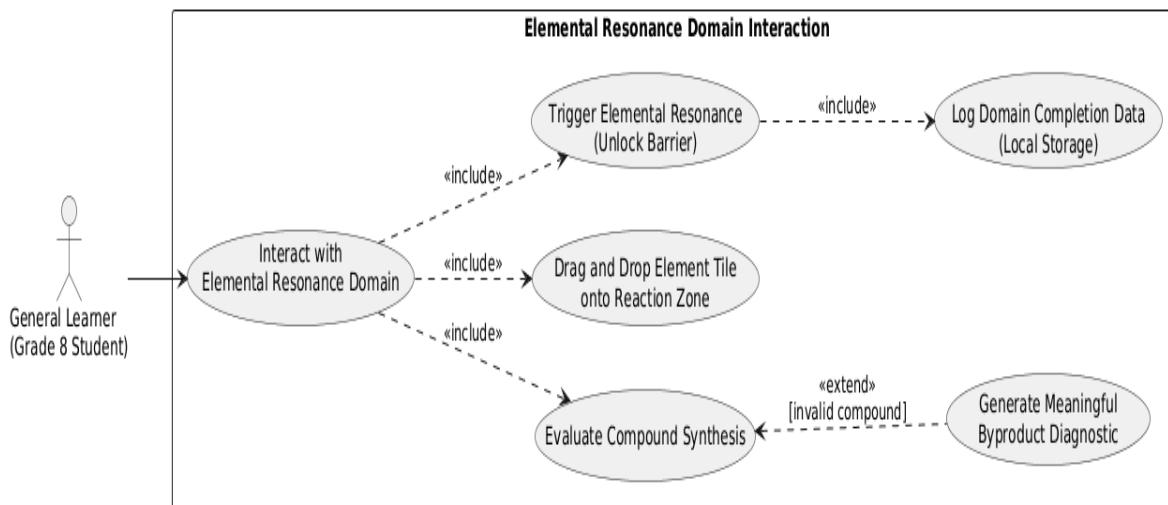
The single-player modules (Elemental Resonance Puzzle Module and Assessment and Progress Module) require internet connectivity exclusively for reading and writing session data to Supabase. The Opponent-Based Challenge Module relies on Supabase Realtime for low-latency bidirectional communication between two competing players. A minimum network bandwidth of 1 Mbps is recommended to ensure uninterrupted match synchronization. All communications between the frontend and Supabase are conducted over HTTPS. No personally identifiable information is transmitted over the network.

3.2. Functional requirements

Module 1: Elemental Resonance Puzzle Module

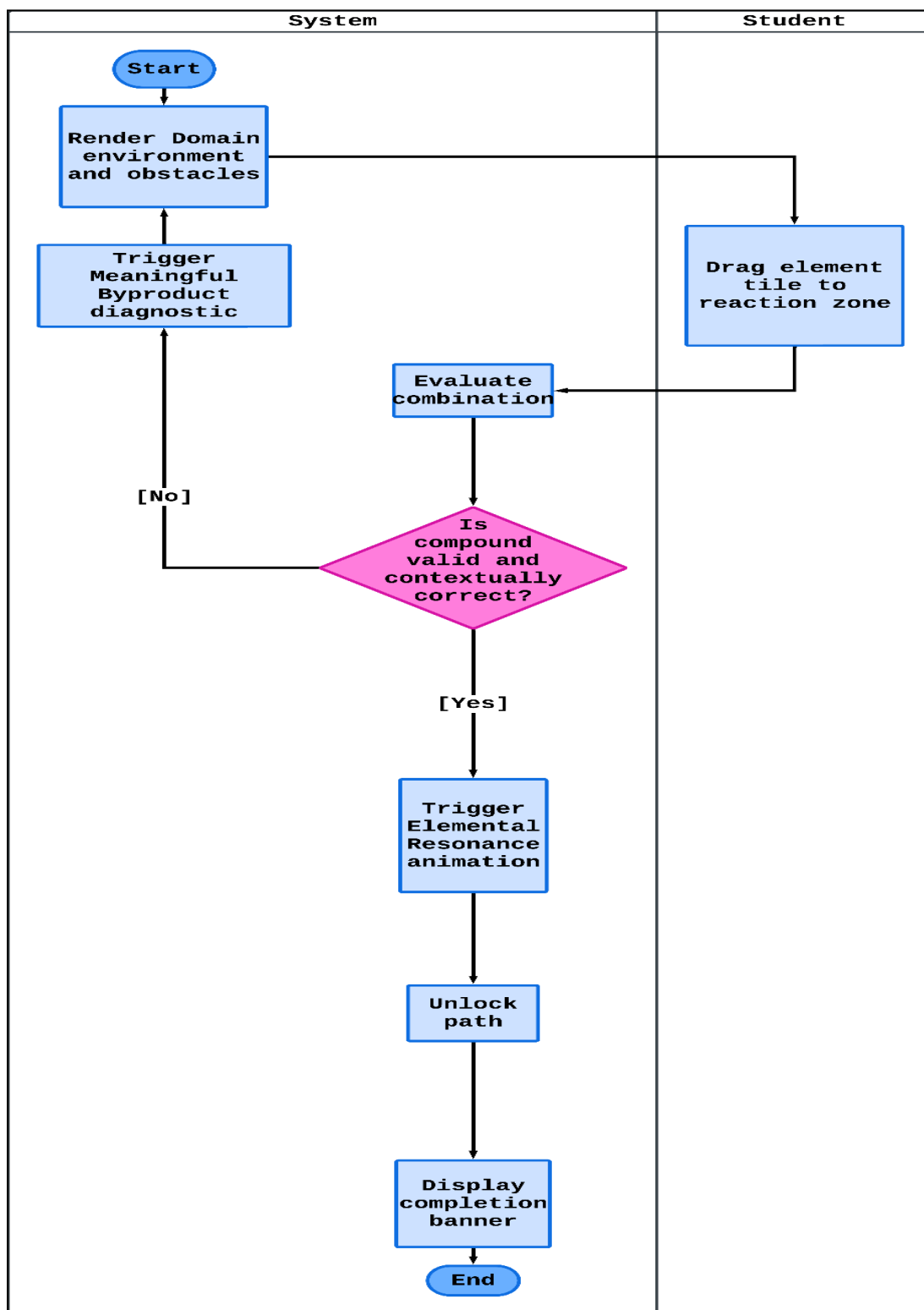
1.1 Elemental Resonance Domain Interaction

- **Use Case Diagram**



- **Use Case Description:** The student interacts with a text-free Elemental Resonance Domain environment. Rather than answering a quiz question, the student drags chemical element tiles onto the reaction zone to synthesize compounds and overcome a physical obstacle. A valid compound synthesis triggers a visual Elemental Resonance effect where the obstacle is cleared and the path forward is unlocked. An invalid combination generates a Meaningful Byproduct diagnostic message for a predefined set of wrong combinations, explaining why the attempt failed based on the chemical properties of the elements involved. Session completion data is logged to Supabase upon clearing the Domain.

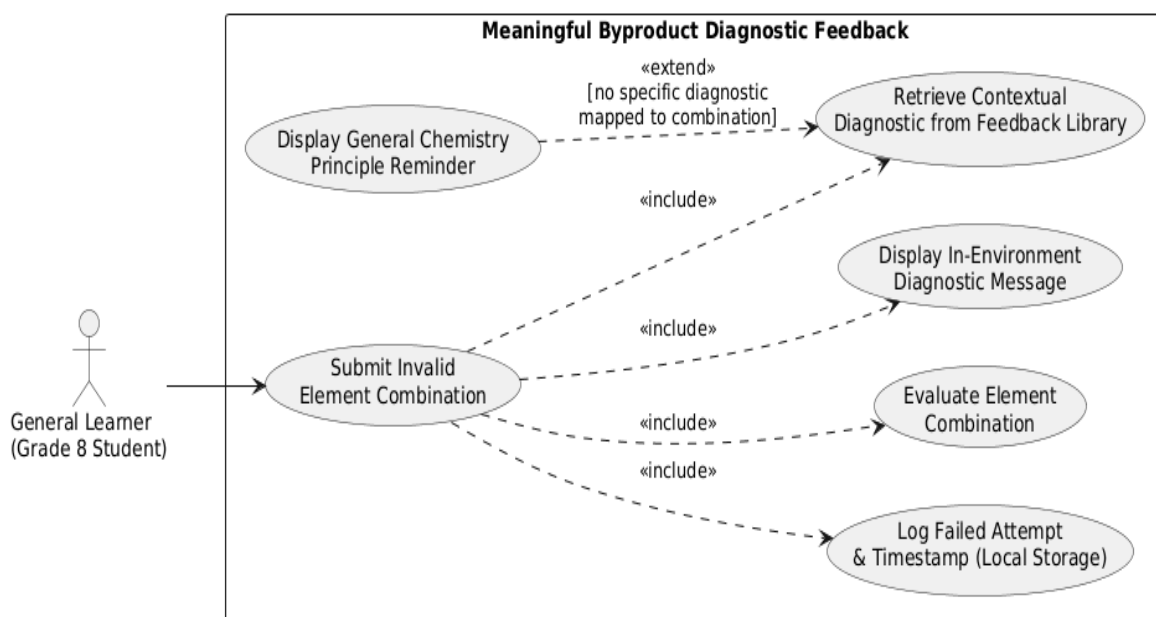
- **Activity Diagram: Elemental Resonance Domain Interaction**



- Wireframe

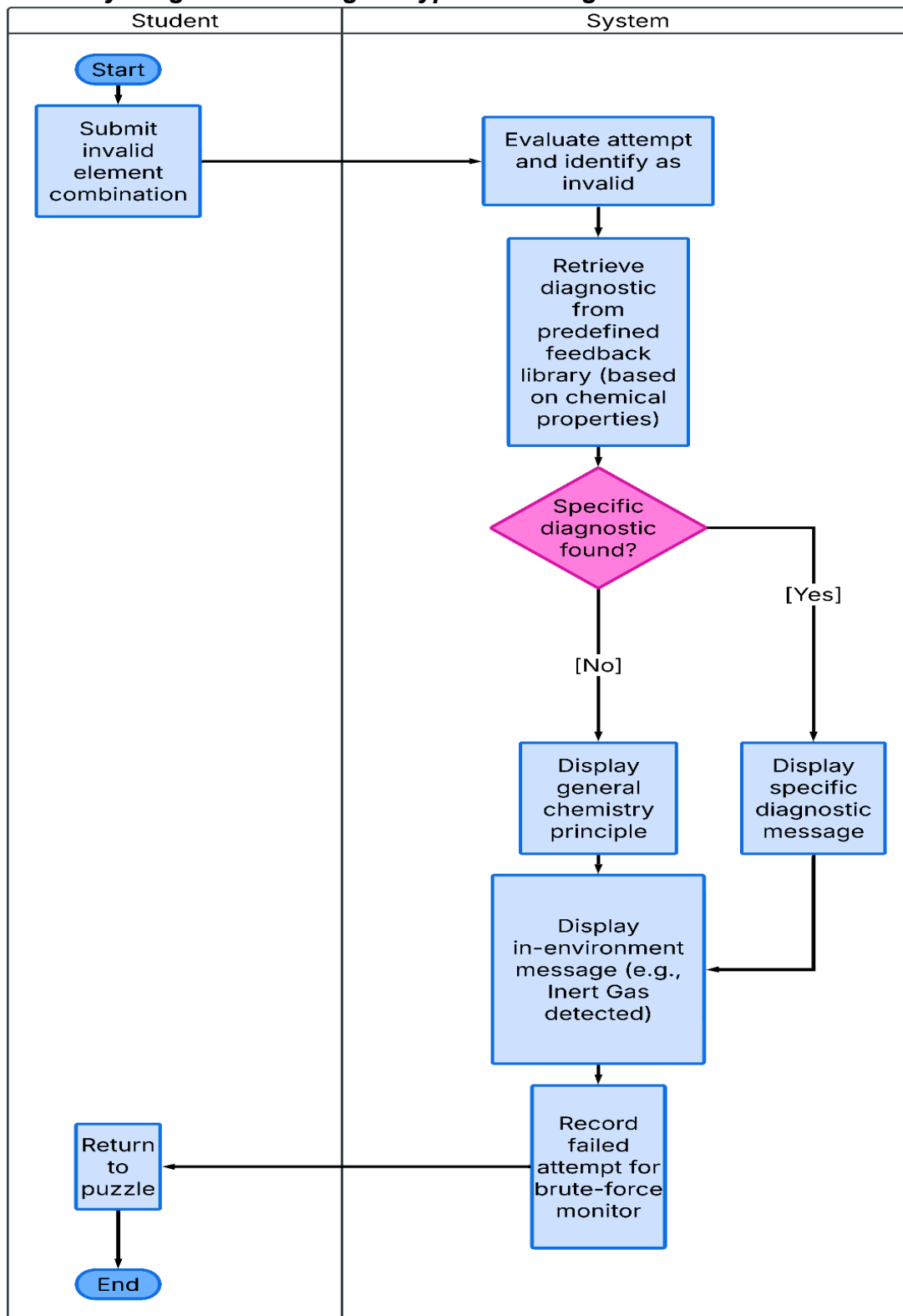
1.2 Meaningful Byproduct Diagnostic Feedback

- Use Case Diagram



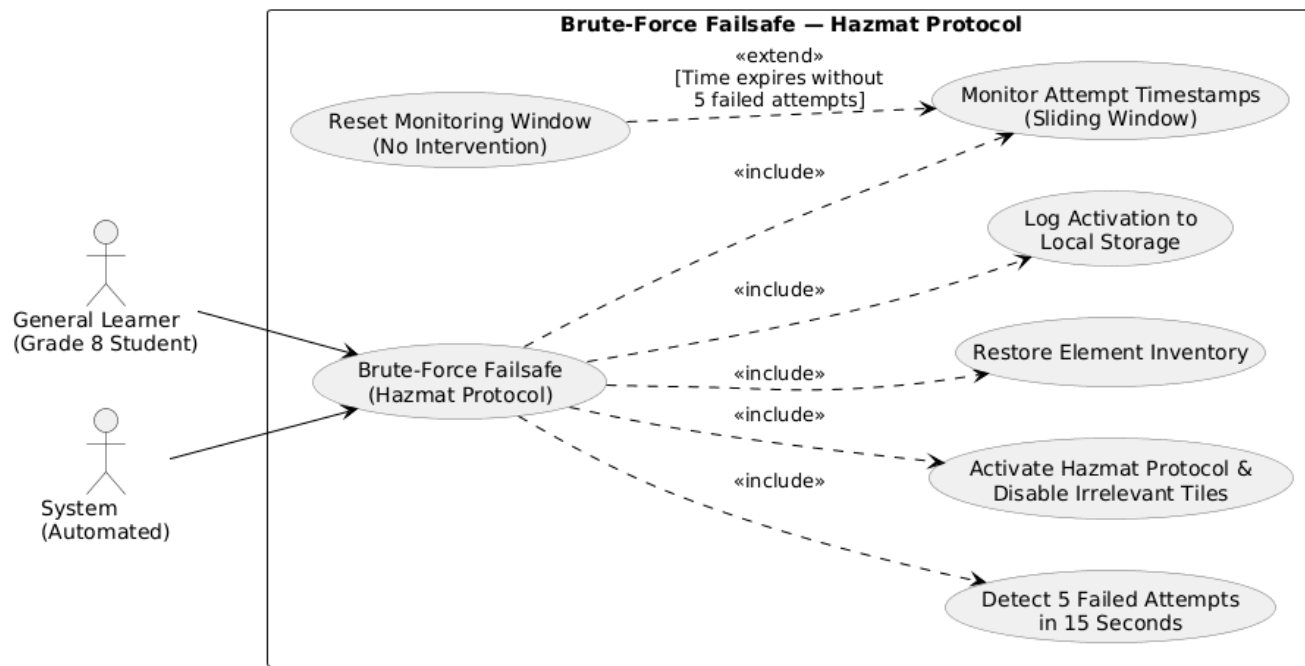
- **Use Case Description:** Upon an invalid compound combination attempt, the system checks the attempted combination against a predefined feedback library. If a specific diagnostic is mapped to that combination, the system displays a contextual, chemistry-based message explaining why the attempt failed based on the actual chemical properties of the elements involved. If no specific diagnostic is mapped, the system displays a general chemistry principle reminder. The failed attempt is recorded to support the Hazmat Protocol monitoring system.

• **Activity Diagram: Meaningful Byproduct Diagnostic Feedback**



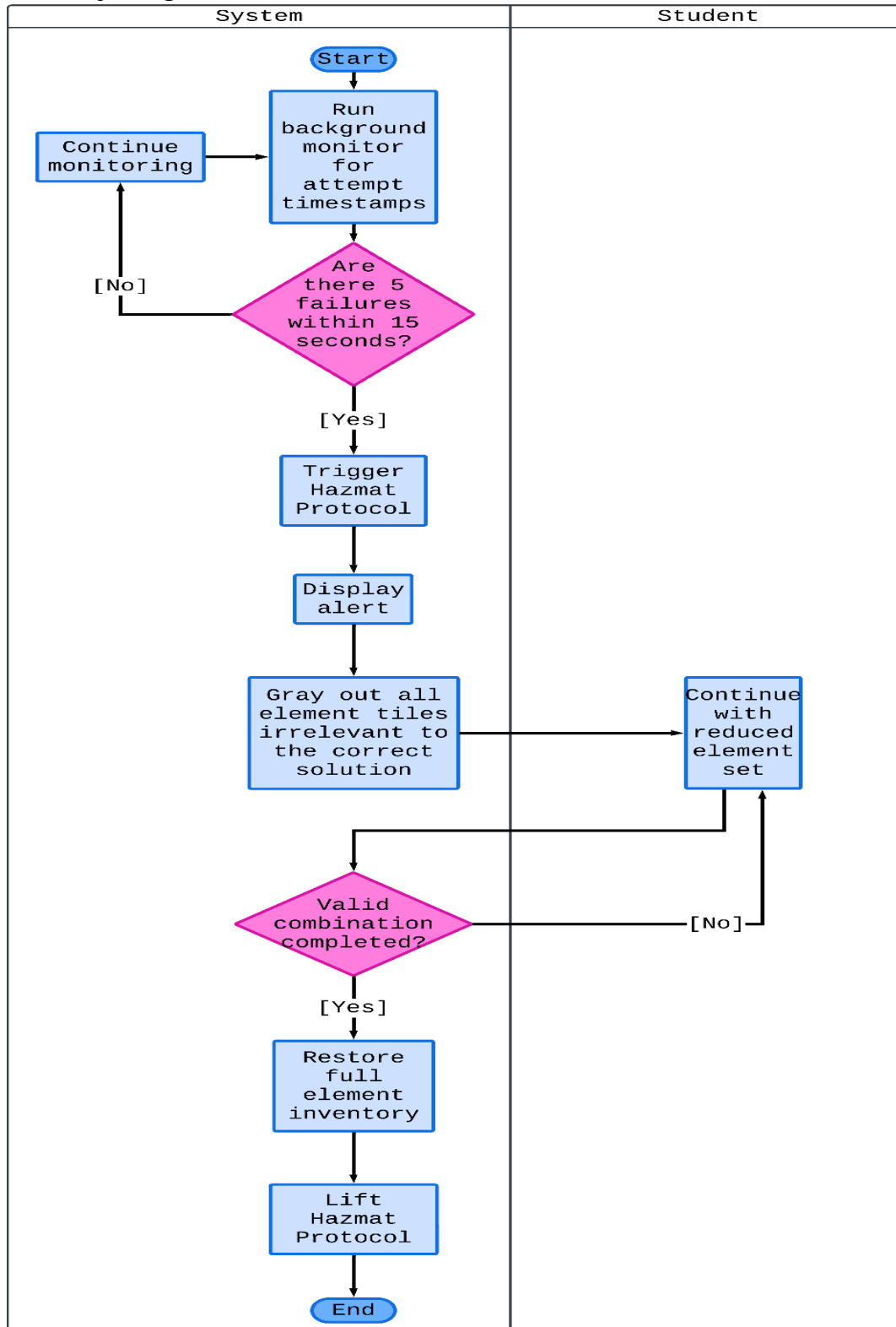
- **Wireframe**

1.3 Brute-Force Failsafe



- **Use Case Diagram**
- **Use Case Description:** When the system detects five or more failed combination attempts within a 15-second sliding window, it automatically activates the Hazmat Protocol. This intervention visually disables element tiles that are irrelevant to the current puzzle, narrowing the student's focus without revealing the correct answer. Once the student successfully synthesizes a valid compound, the full element inventory is restored and the Hazmat Protocol is lifted. The activation is recorded to Supabase as part of the session performance log.

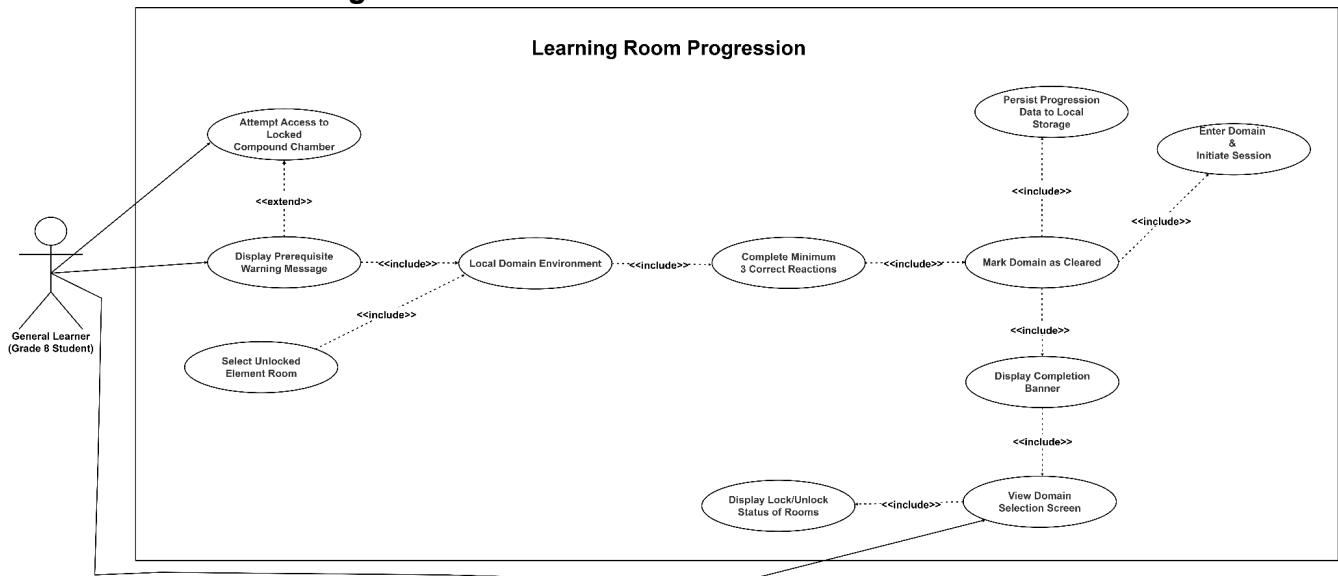
- Activity Diagram: Brute-Force Failsafe



- **Wireframe**

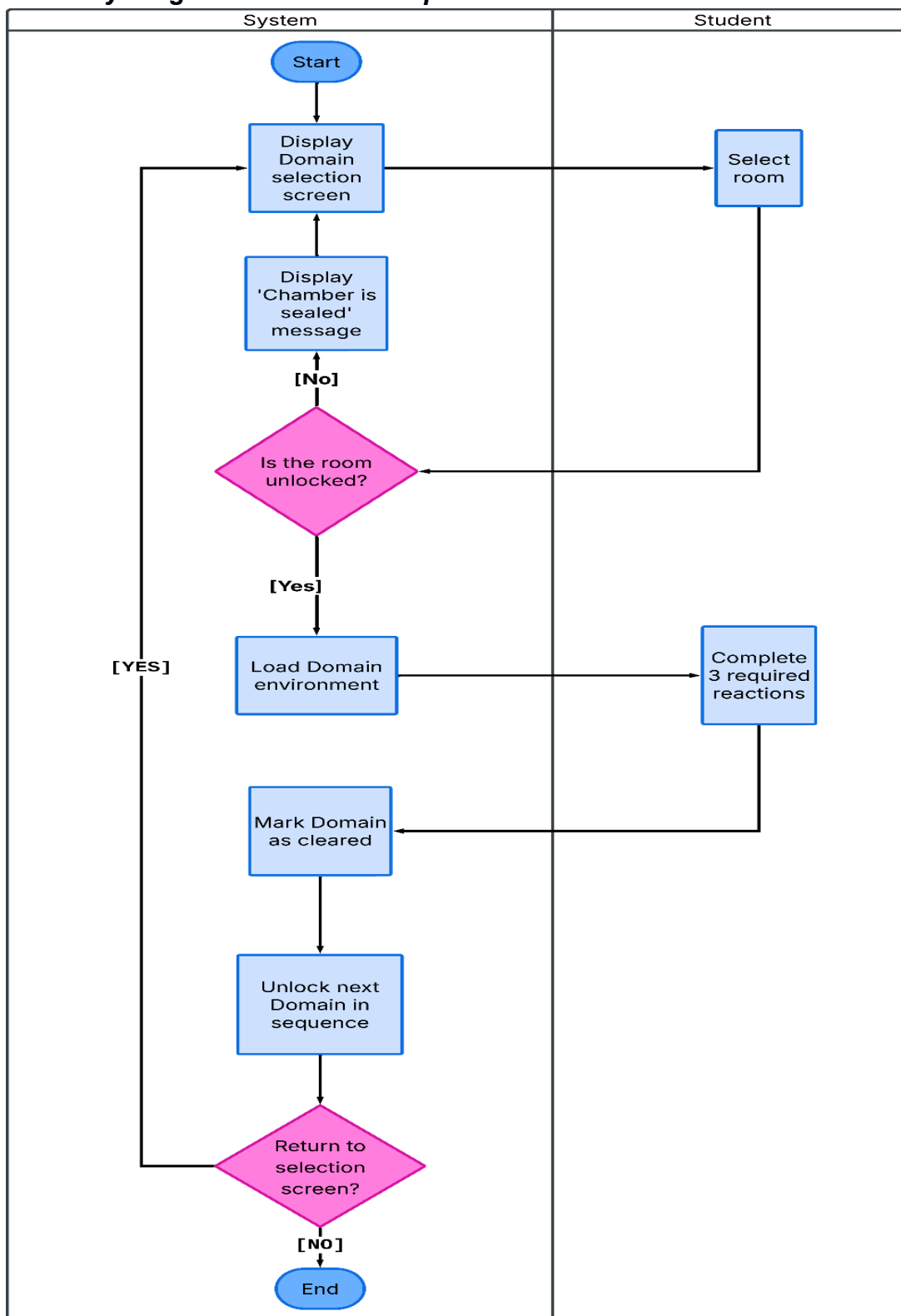
1.4 Learning Room Progression

- **Use Case Diagram**



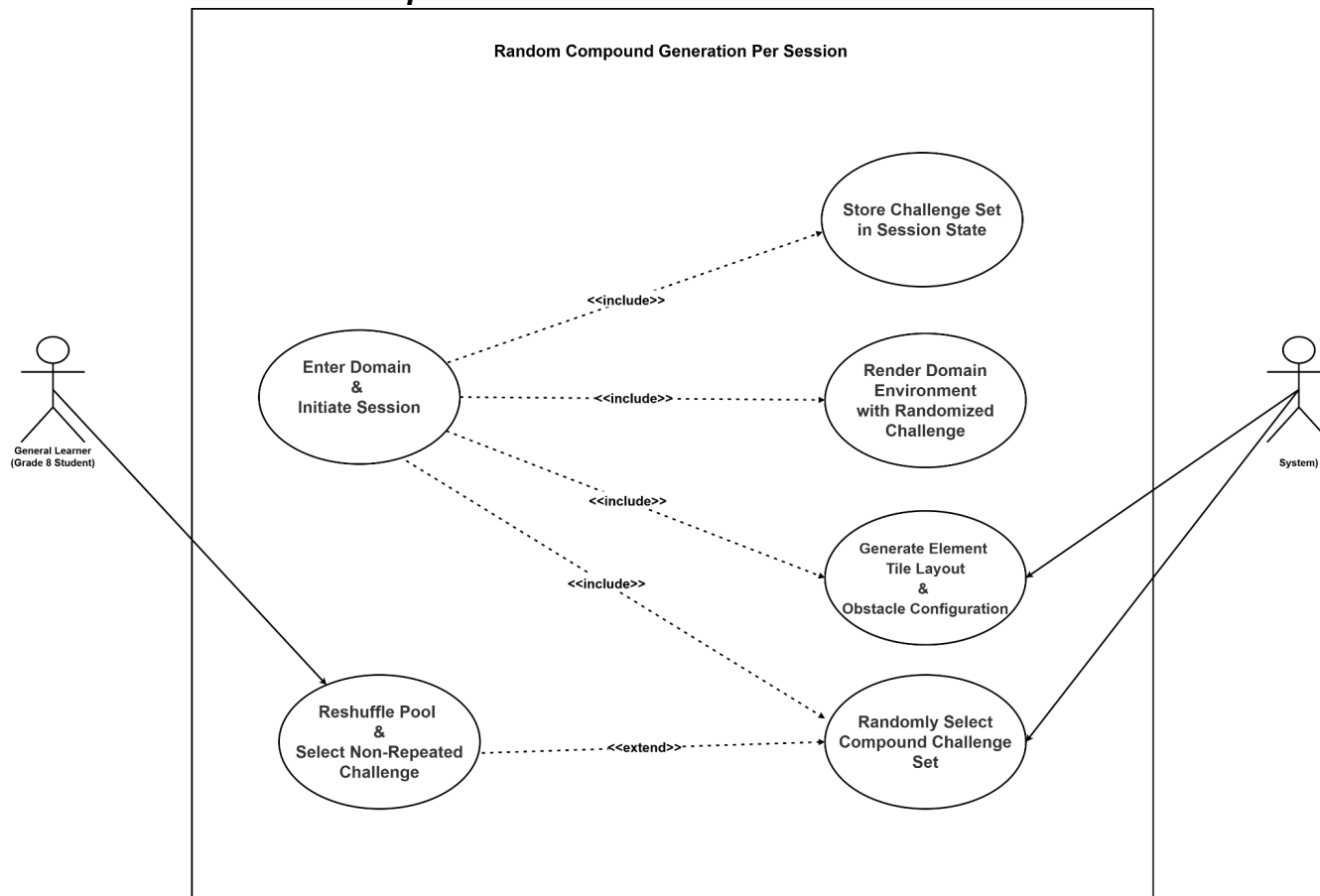
- **Use Case Description:** The system enforces a sequential unlocking structure for Domains. Students must clear foundational Element Rooms before accessing advanced Compound Chambers. A Domain is considered cleared when the student completes a minimum of three correct reactions within it. Upon clearing a Domain, the system marks it as complete, unlocks the next Domain in sequence, and displays a completion banner. Progression data is logged to Supabase so the student's unlock status persists across sessions.

- Activity Diagram: Random Compound Generation Per Session



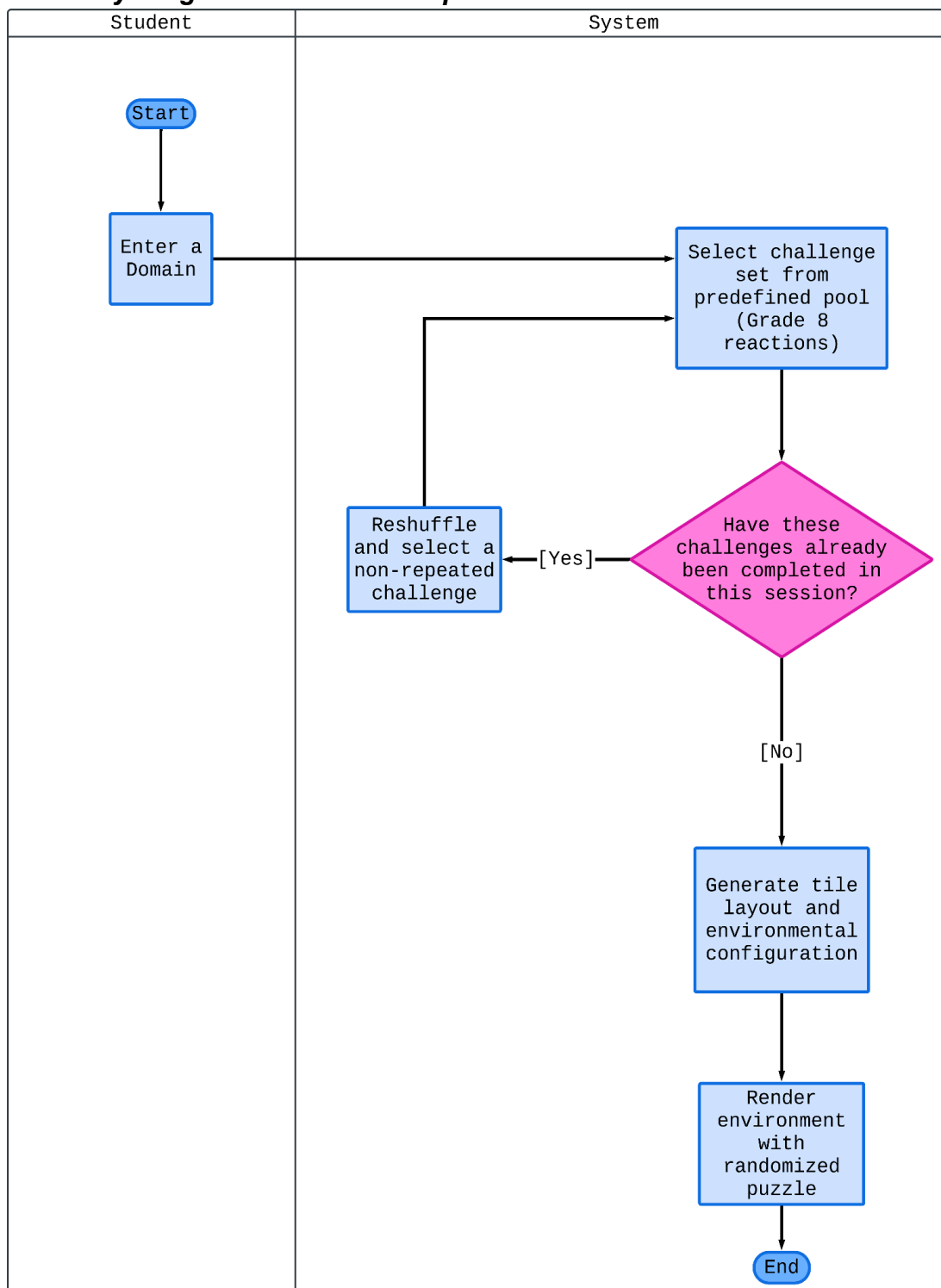
- **Wireframe**

1.5 Random Compound Generation Per Session



- **Use Case Diagram**
- **Use Case Description:** At the start of each Domain session, the system automatically selects a randomized set of chemical compound challenges from a predefined pool of high school chemistry reactions. The system checks whether the selected challenges have already been completed in the current session and reshuffles if necessary, ensuring students cannot rely on memorizing a fixed sequence of answers across sessions. The randomized challenge set is stored in session state for the duration of the Domain.

- **Activity Diagram: Random Compound Generation Per Session**

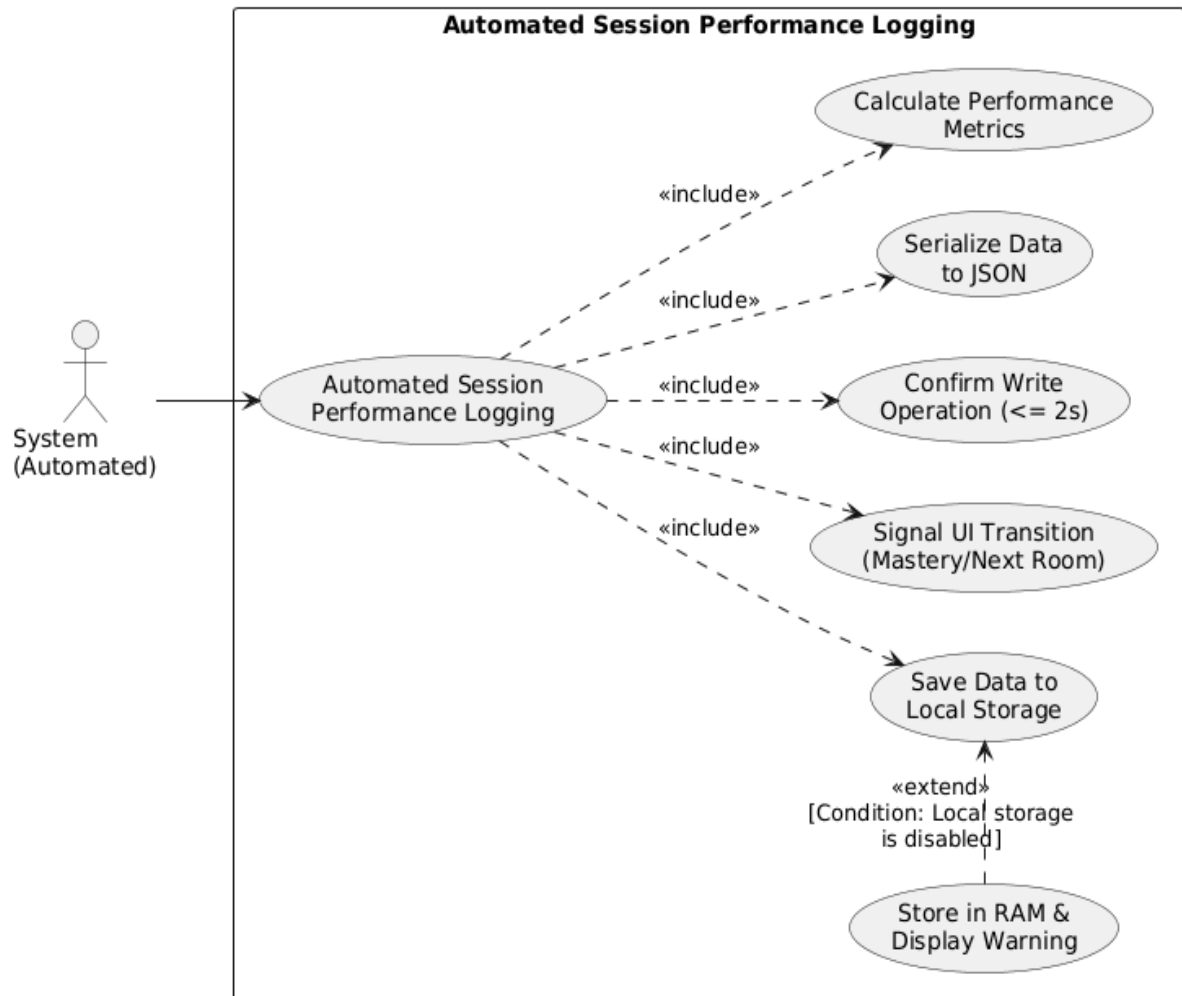


- **Wireframe**

Module 2: Local Assessment & Progress Module

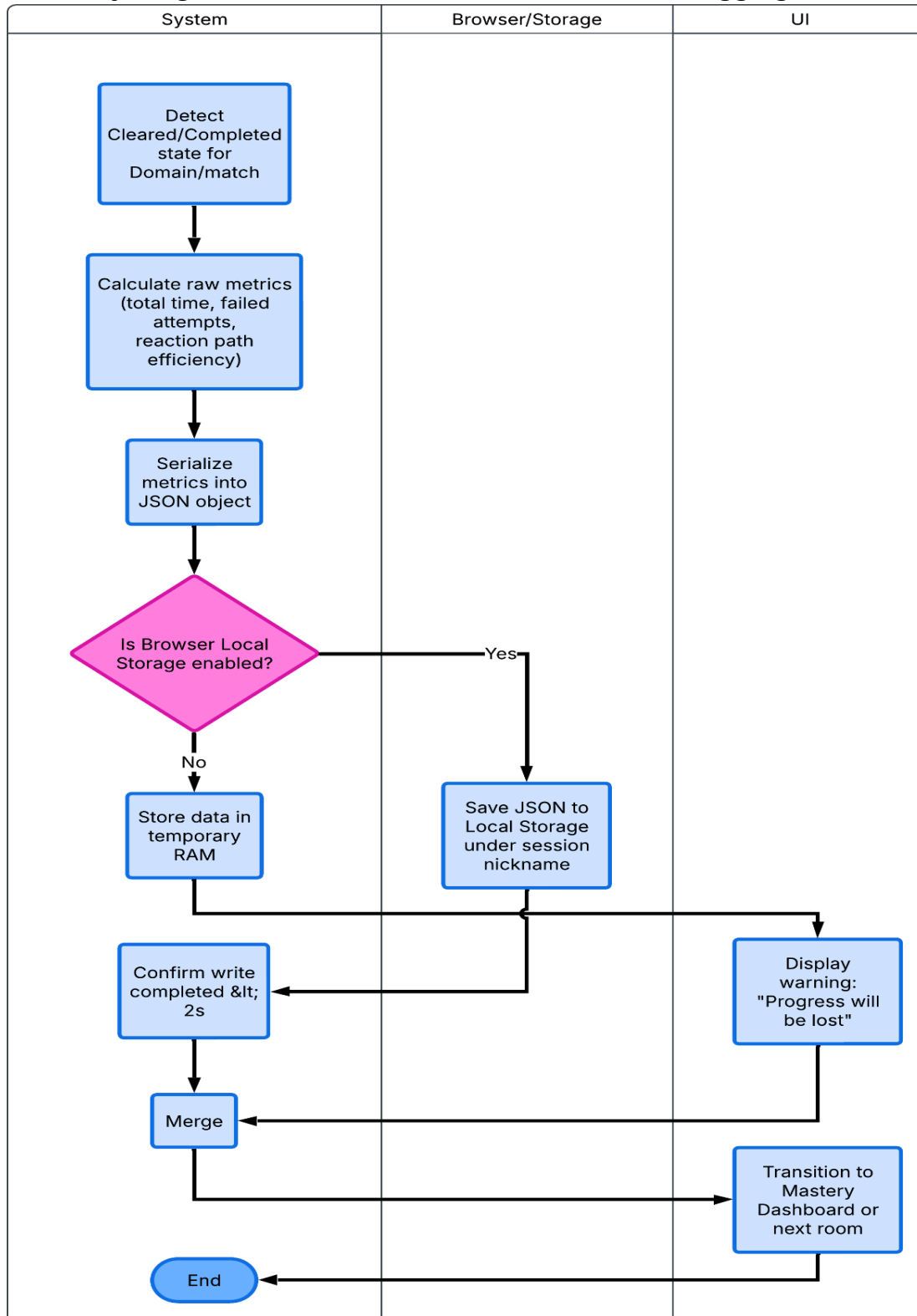
2.1 Automated Session Performance Logging

- **Use Case Diagram**



- **Use Case Description:** Upon completion of any learning Domain, the system automatically calculates the student's performance metrics based on completion speed and compound accuracy, then writes this data to Supabase under the student's temporary session nickname. The write operation is completed within less than 2 seconds. If the Supabase write fails, the system displays a user-friendly error message without crashing. The UI then transitions to the Mastery Dashboard or the next available Domain.

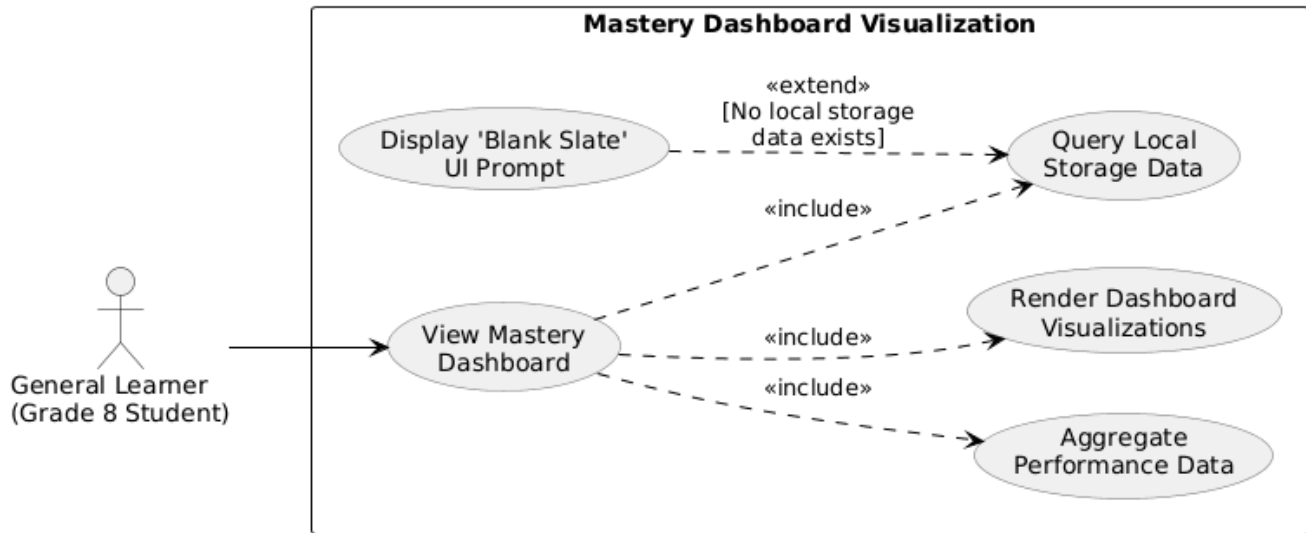
• **Activity Diagram: Automated Session Performance Logging**



- **Wireframe**

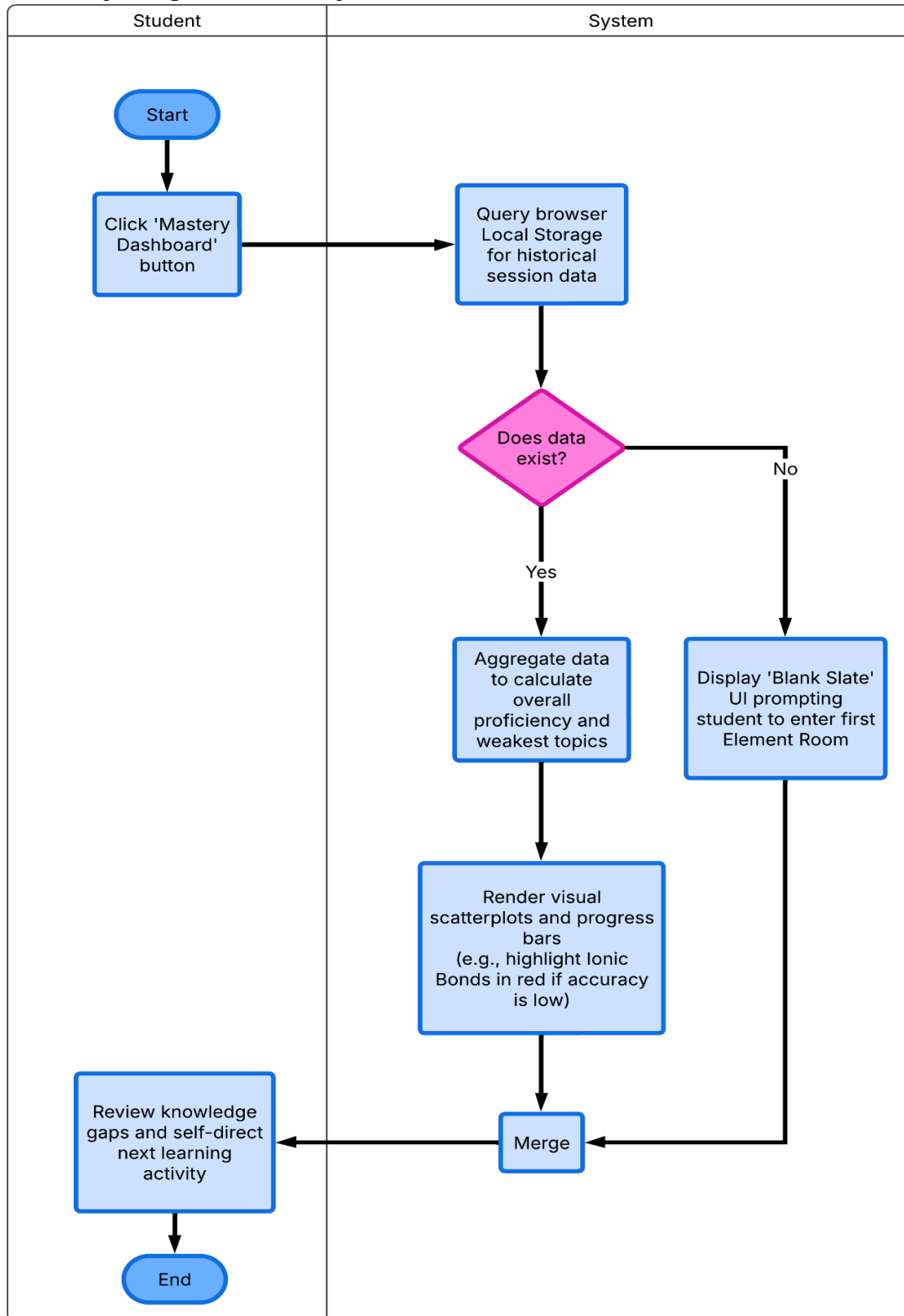
2.2 Mastery Dashboard Visualization

- **Use Case Diagram**



- **Use Case Description:** The student accesses their personalized Mastery Dashboard, which queries Supabase for their historical session data and renders it as visual progress indicators per Domain. If no session data exists, the system displays a prompt encouraging the student to complete their first Domain. The dashboard highlights which Domains have low accuracy, allowing the student to independently identify knowledge gaps without teacher intervention.

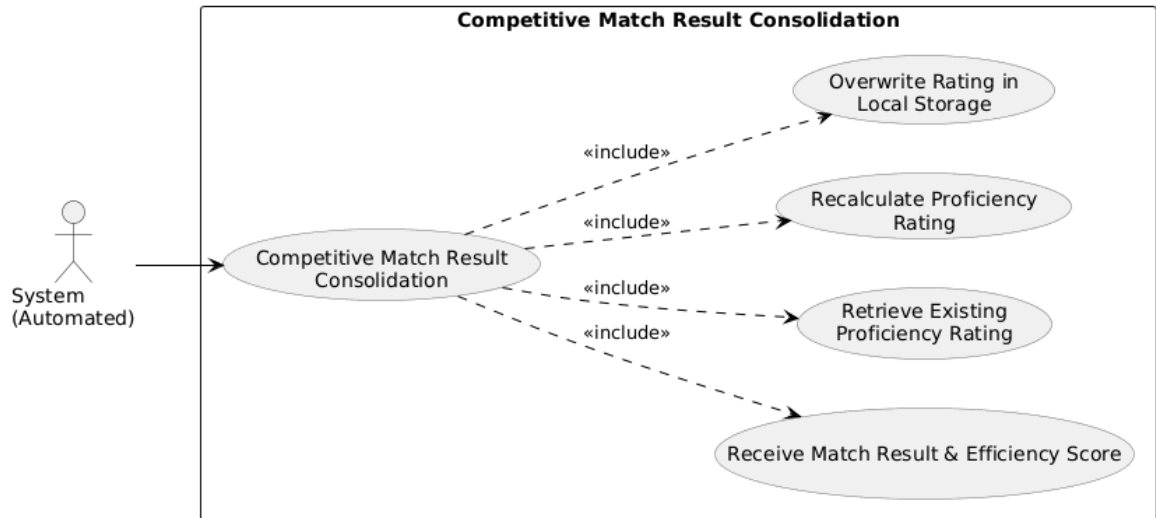
- **Activity Diagram: Mastery Dashboard Visualization**



- **Wireframe**

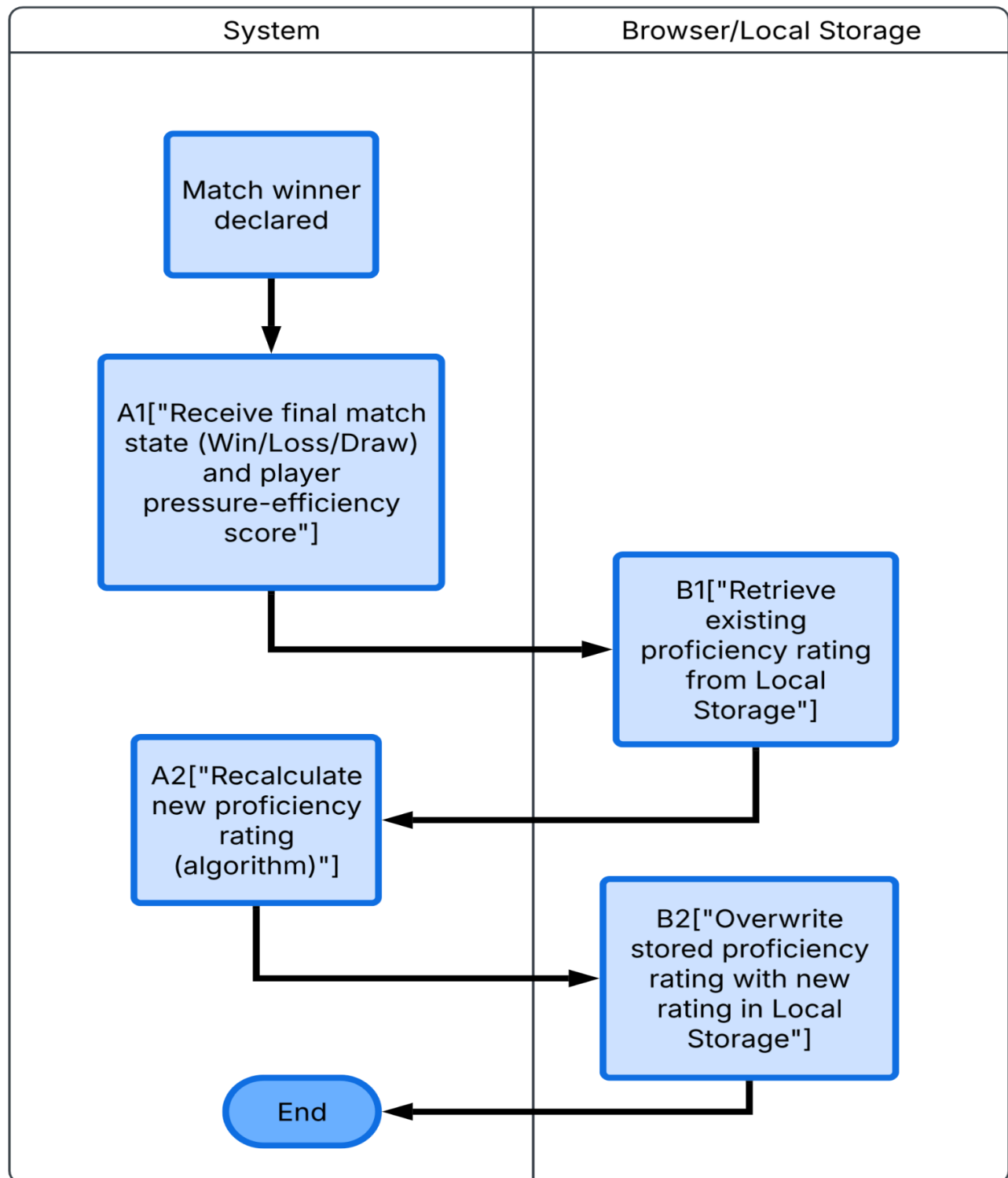
2.3 Competitive Match Result Consolidation

- **Use Case Diagram**



- **Use Case Description:** After a multiplayer match concludes, the system retrieves the student's existing proficiency data from Supabase, recalculates their overall rating based on the match result and Speed-to-Compound efficiency score, and overwrites the stored rating in Supabase. The updated data is reflected in the student's Mastery Dashboard.

- **Activity Diagram: Competitive Match Result Consolidation**

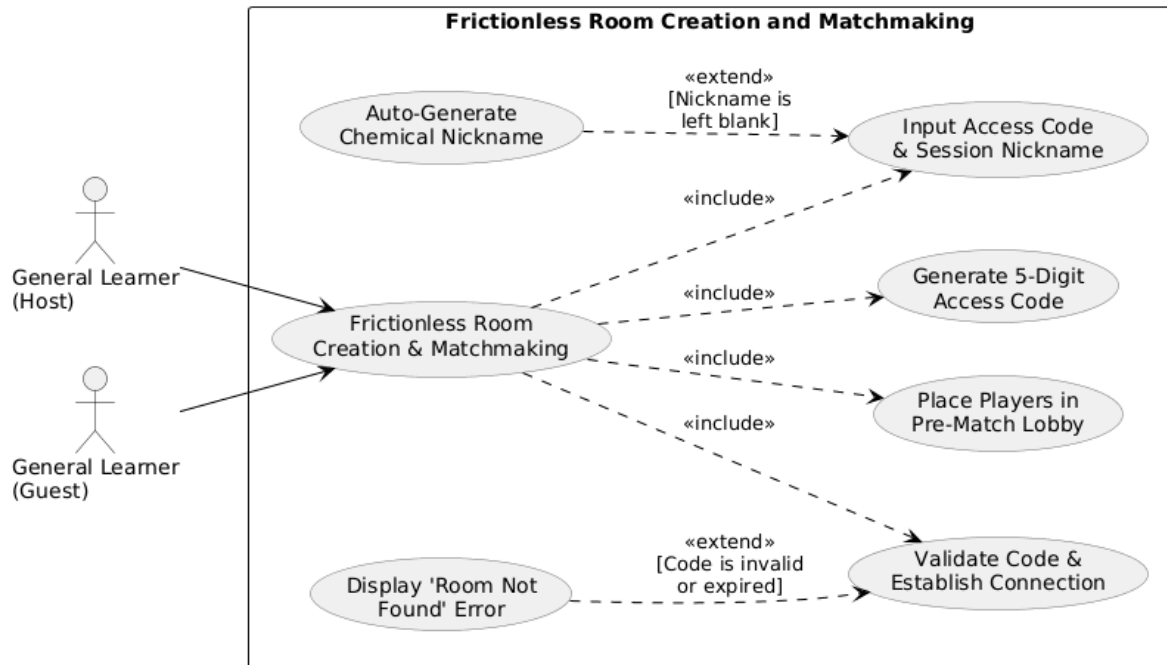


- **Wireframe**

Module 3: Opponent-Based Challenge Module

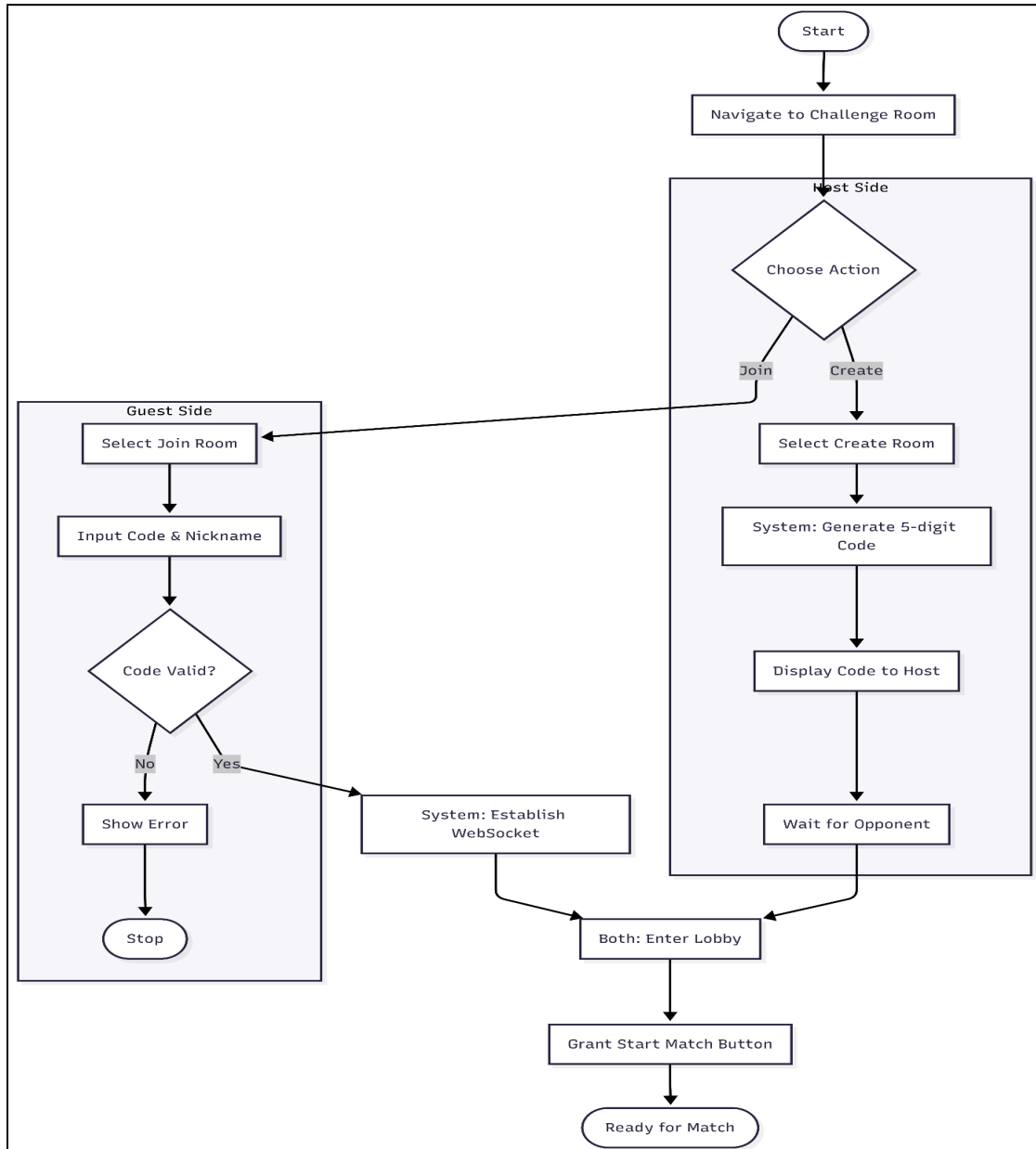
3.1 Room Creation and Matchmaking

- **Use Case Diagram**



- **Use Case Description:** A host student navigates to the Opponent-Based Challenge section and selects to create a room. The system auto-generates a 5-digit access code and displays it to the host. The host waits in a pre-match lobby. A guest student enters the same access code along with a temporary session nickname to join the room. The system validates the code, establishes a Supabase Realtime connection between both players, and places them in the pre-match lobby. If the code is invalid or expired, the system displays a room not found error. Once both players are connected, the host can start the match.

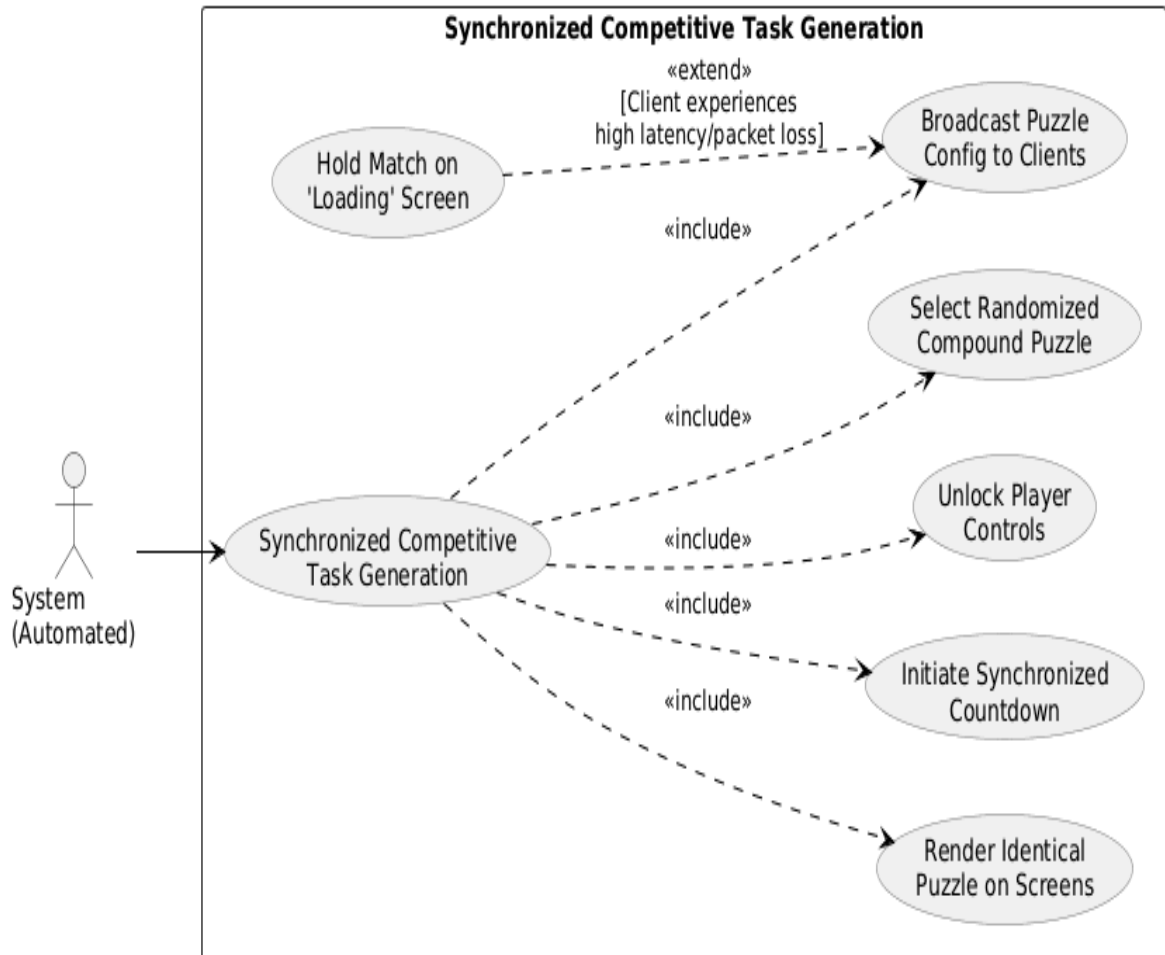
- Activity Diagram



- Wireframe

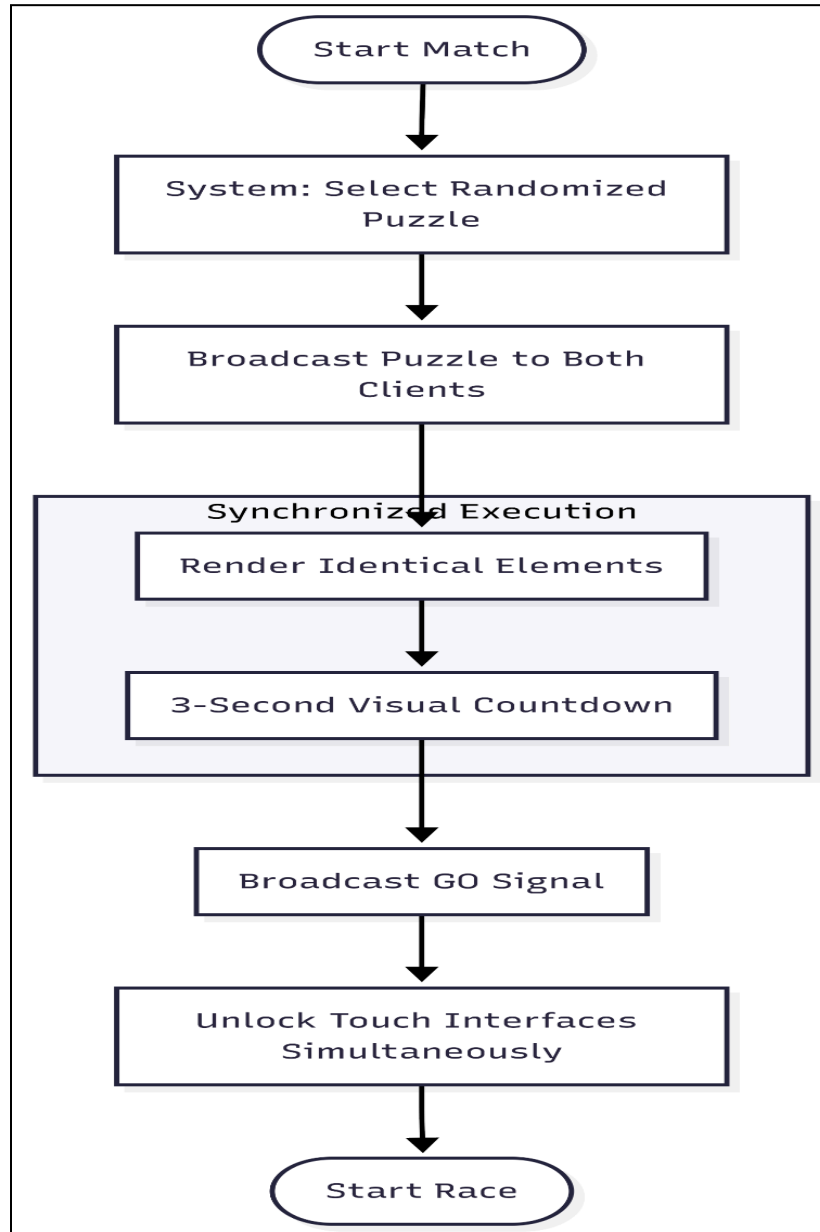
3.2 Synchronized Competitive Task Generation

- **Use Case Diagram**



- **Use Case Description:** Once the match is initiated, the system automatically selects a randomized compound challenge from the predefined pool and broadcasts the same puzzle configuration to both connected players simultaneously via Supabase Realtime. Both players see identical element inventories and obstacles. A 3-second visual countdown is displayed before player controls are unlocked and the race begins.

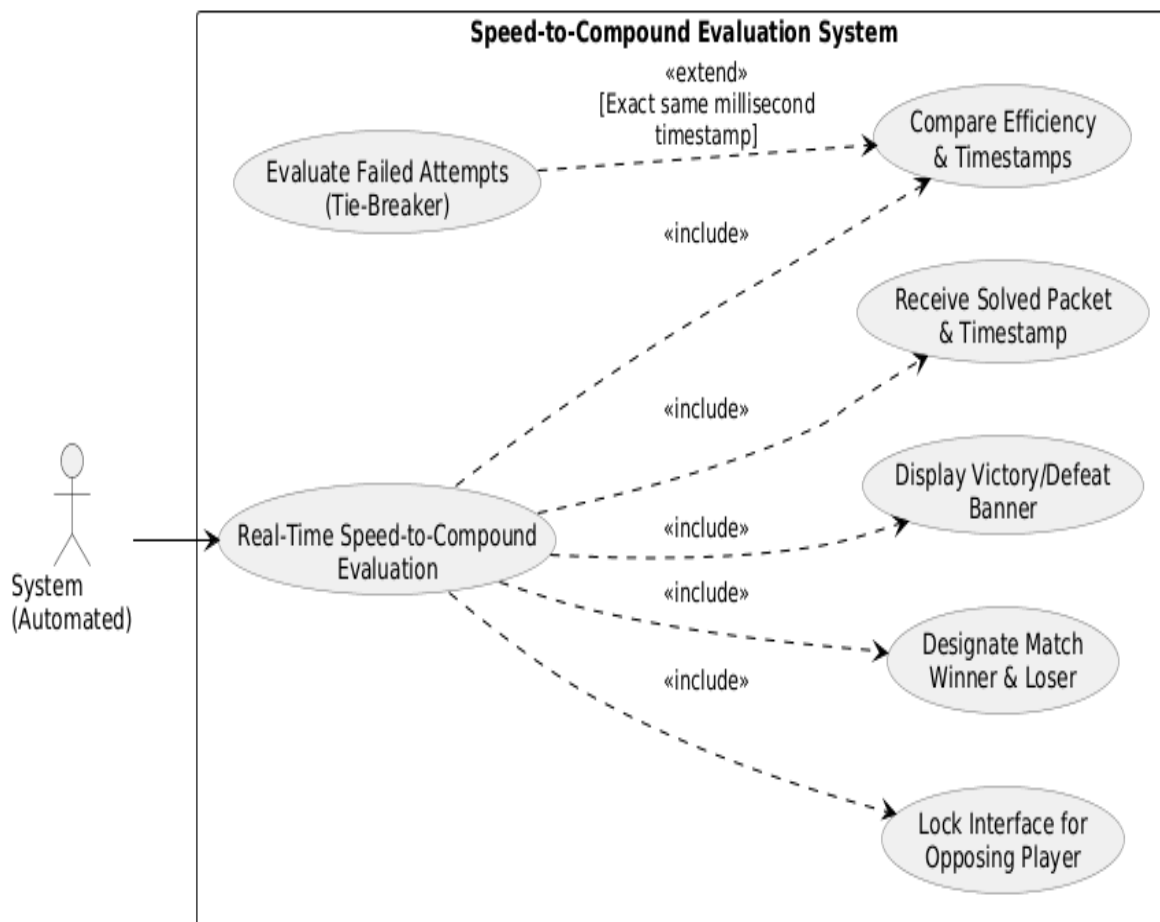
- **Activity Diagram**



- **Wireframe**

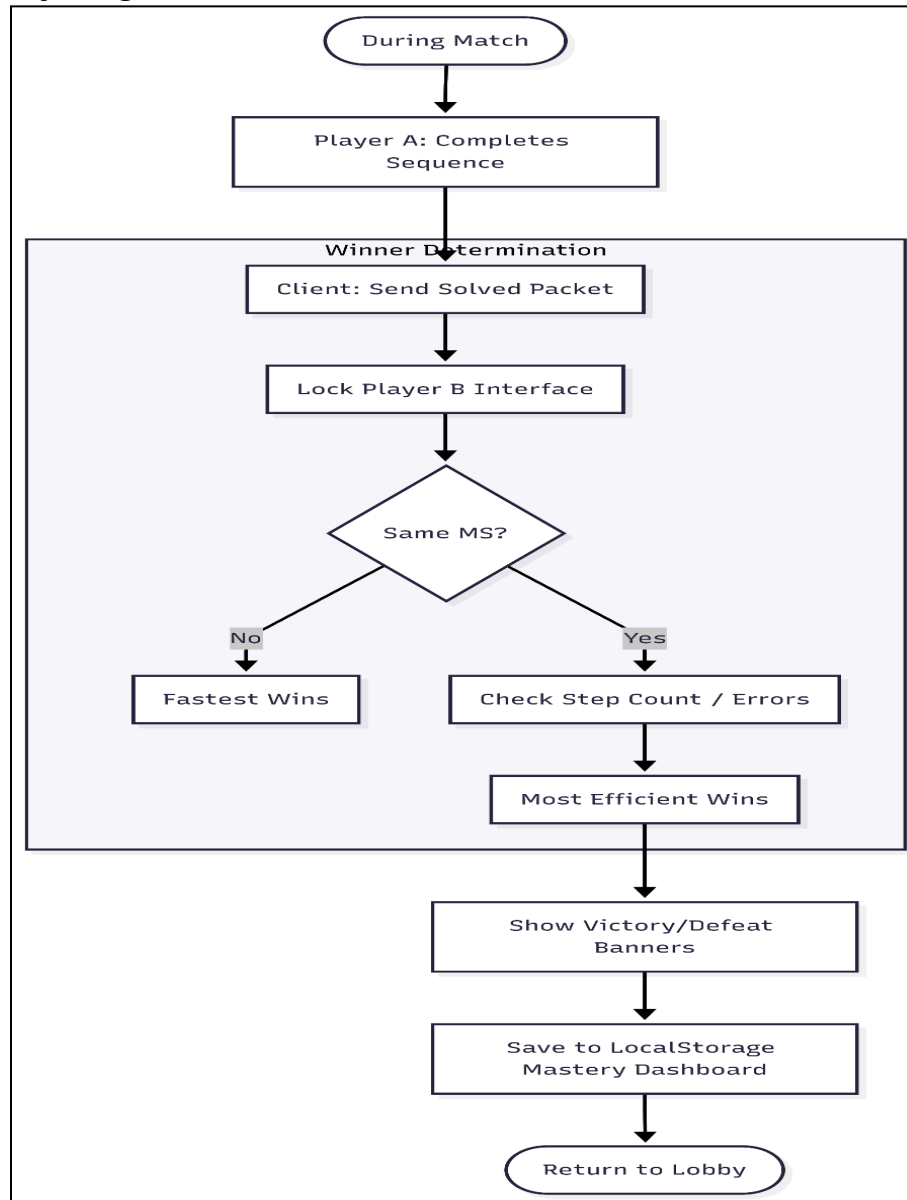
3.3 Real-Time Speed-to-Compound Evaluation

- **Use Case Diagram**



- **Use Case Description:** During the match, both players attempt to synthesize the correct compound. The first player to complete a valid synthesis sends a solved signal to the system via Supabase Realtime. The system locks the opposing player's interface, determines the winner, and displays victory and defeat banners to the respective players. If both players solve at the exact same millisecond, the player with fewer failed attempts is declared the winner. Match results and efficiency scores are saved to Supabase and reflected in both players' Mastery Dashboards.

- **Use Case Description:** During a match, the system dynamically compares the input speeds and reaction path efficiencies of both players. The system determines the match winner with 100% computational accuracy based on who reaches the stable compound first using the most efficient path.
- **Activity Diagram**



- **Wireframe**

3.4 Non-functional requirements

Performance

- *Session performance data must be logged to Supabase within less than 2 seconds per Domain completion.*
- *The Opponent-Based Challenge Module must establish a Supabase Realtime connection in less than 5 seconds after a valid 5-digit access code is entered.*
- *The Mastery Dashboard must render the student's latest session data within 3 seconds of being accessed.*
- *The drag-and-drop puzzle interface must respond to user input with no noticeable delay on standard devices.*
- *The system must support at least 2 concurrent users per competitive room without degradation in real-time synchronization.*

Security

- *The system must not collect, transmit, or store personally identifiable information. Users are identified only by a self-assigned temporary nickname.*
- *All communication between the frontend and Supabase must be conducted over HTTPS.*
- *The Opponent-Based Challenge Module must validate access codes server-side before permitting room entry.*
- *The system does not implement any authentication mechanism. No credentials are stored or transmitted.*

Reliability

- *The single-player modules must remain functional as long as Supabase is accessible.*
- *In the event of a Supabase connection failure during a competitive match, the system must display a user-friendly error message and terminate the session gracefully without data corruption.*
- *Session performance data written to Supabase must persist across browser restarts and device switches as long as the student uses the same session nickname.*

Usability

- *The system must be fully mobile-responsive, supporting touch-based interaction across devices with screen widths from 360px to 1920px.*
- *All puzzle interactions must be text-free, relying exclusively on visual drag-and-drop*

mechanics.

- *The system must not require user registration or authentication to access any module.*

Compatibility

- *The system must be compatible with Google Chrome v110+, Mozilla Firefox v100+, and Safari v15+.*
- *The system must support Supabase Realtime as a required dependency for the Opponent-Based Challenge Module and session data persistence.*